

Complex Analysis

5.6 Taylor Series (Analytic Series)

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \Rightarrow f(z) = a_0 + a_1 (z - z_0) + a_2 (z - z_0)^2 + \dots$$

or

$$f(z) = \sum_{n=0}^{\infty} \frac{f^n(z)}{n!} (z - z_0)^n \text{ (at } z = z_0 \text{)}$$

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!} (z - z_0)^2 + \dots$$

$$a_0 = f(z_0), a_1 = f'(z_0), a_2 = \frac{f''(z_0)}{2!}$$

e^z , $\sin z$, $\cos z$ & $\log z$ all are Taylor's series about origin and known as Maclaurin series.

Example: Find Taylor's series of following functions.

(a) $f(z) = \sin z$ at $z_0 = \frac{\pi}{4}$.

Solution: $f(z_0) = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}, \quad f'(z_0) = \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$f''(z_0) = -\sin \frac{\pi}{4} = -\frac{1}{\sqrt{2}}, \quad f'''(z_0) = -\cos \frac{\pi}{4} = -\frac{1}{\sqrt{2}}$

$f(z) = f(z_0) + (z - z_0)f'(z_0) + \frac{(z - z_0)^2}{2!}f''(z_0) + \dots$

$f(z) = \frac{1}{\sqrt{2}} + \left(z - \frac{\pi}{4}\right)\frac{1}{\sqrt{2}} + \left(z - \frac{\pi}{4}\right)\frac{1}{2!}\left(-\frac{1}{\sqrt{2}}\right) + \left(z - \frac{\pi}{4}\right)^3\frac{1}{3!}\left(-\frac{1}{\sqrt{2}}\right) + \dots$

(b) $f(z) = \log \cosh z$ at $z_0 = 0$.

Solution: $f(z) = \log \cosh z, \quad f(z_0) = \log 1 = 0,$

$f'(z) = \frac{\sinh z}{\cosh z} = \tanh z, \quad f'(z_0) = 0 \quad f''(z) = \sec^2 z, \quad f''(z_0) = 1$

$f'''(z) = 2 \sec^2 z \tanh z, \quad f'''(z_0) = 0$

$f^{iv}(z) = -2(\tanh^2 z \operatorname{sech}^2 z + \sec^2 z \cdot \operatorname{sech}^2 z), \quad f^{iv}(z_0) = -2$

$f(z) = 0 + (z - 0) \cdot 0 + \frac{(z - 0)^2}{2!} \cdot 1 + \frac{(z - 0)^3}{3!} \cdot 0 + \frac{(z - 0)^4}{4!} \times (-2) + \dots$

$f(z) = \frac{z^2}{2!} - \frac{2z^4}{4!} + \dots$

(c) $f(z) = \frac{1}{1-z}, \quad |z| < 1 \text{ \& } z_0 = 0$

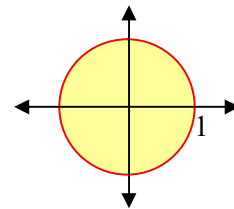
Solution: $f(z) = \frac{1}{1-(z-0)} = (1-z)^{-1} = 1 + z + z^2 + z^3 + \dots$

Or

$f(z_0) = \frac{1}{1-0} = 1, \quad f'(z) = \frac{1}{(1-z)^2}, \quad f'(z_0) = 1$

$f''(z) = \frac{2}{(1-z)^3}, \quad f''(z_0) = 2$

$f(z) = f(z_0) + (z - z_0)f'(z_0) + \frac{(z - z_0)^2}{2!}f''(z_0) + \dots$



$$= 1 + z + z^2 + z^3 + \dots$$

(d) $f(z) = \frac{1}{1-z}, \quad |z| < 1 \text{ \& } z_0 = i$

Solution: $f(z) = \frac{1}{(1-i)-(z-i)} = \frac{1}{(1-i)\left[1-\frac{(z-i)}{(1-i)}\right]}$

$$f(z) = \frac{1}{1-i} \left[1 - \frac{z-i}{1-i} \right]^{-1} = \frac{1}{1-i} \left[1 + \frac{z-i}{1-i} + \left(\frac{z-i}{1-i} \right)^2 + \dots \right]$$

$$R_m = \lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}}$$

$$f(z) = \left[\frac{1}{1-i} + \frac{(z-i)}{(1-i)^2} + \frac{(z-i)^2}{(1-i)^3} + \dots \right]$$

$$a_{n+1} = \frac{1}{(1-i)^{n+2}}, \quad a_n = \frac{1}{(1-i)^{n+1}}, \quad \frac{a_n}{a_{n+1}} = \frac{1}{(1-i)^{n+1}} \times \frac{(1-i)^{n+2}}{1}$$

$$\frac{a_n}{a_{n+1}} = (1-i) = \sqrt{2} = R$$

(e) $f(z) = \tanh z$ at $z_0 = 0$.

Solution: $f(z) = \tanh z, \quad f(z_0) = 0, \quad f'(z) = \sec^2 z, \quad f'(z_0) = 1$

$$f'(z) = 1 - \tanh^2 z = 1 - [f(z)]^2$$

$$f''(z) = -2f(z)f'(z), \quad f''(z_0) = 0$$

$$f'''(z) = -2[f(z)f''(z) + f'(z)f'(z)] = -2[1] = -2$$

$$f(z) = z - \frac{2z^3}{3!} + \frac{16z^5}{5!} - \dots$$

(e) $f(z) = \frac{1}{(1+z)^2}$ about the point $z = -i$

Solution: $f(z) = \frac{1}{[1+(z+i)-i]^2} = \frac{1}{[(1-i)+(z+i)]^2}$

$$f(z) = \frac{1}{(1-i)^2 \left[1 + \frac{z+i}{1-i}\right]^2} = \frac{1}{(1-i)^2} \left[1 + \frac{z+i}{1-i}\right]^{-2}$$

Binomial expansion:

$$(1 \pm x)^n = 1 \pm nx + \frac{n(n-1)}{2!} x^2 \pm \frac{n(n-1)(n-2)}{3!} x^3 + \dots$$

$$(x+y)^n = x^n \left(1 + \frac{y}{x}\right)^n = x^n \left[1 + \frac{ny}{x} + \frac{n(n-1)}{2!} \frac{y^2}{x^2} + \dots\right]$$

$$f(z) = \frac{1}{(1-i)^2} \left[1 - 2\left(\frac{z+i}{1-i}\right) + \frac{(-2)(-2-1)}{2!} \left(\frac{z+i}{1-i}\right)^2 + \frac{(-2)(-2-1)(-2-2)}{3!} \left(\frac{z+i}{1-i}\right)^3 + \dots\right]$$

$$= \frac{1}{(1-i)^2} \left[1 - 2\left(\frac{z+i}{1-i}\right) + 3\left(\frac{z+i}{1-i}\right)^2 - 4\left(\frac{z+i}{1-i}\right)^3 + \dots\right]$$

$$= \frac{(1+i)^2}{[(1-i)(1+i)]^2} \left[1 - 2\left(\frac{z+i}{1-i}\right) + 3\left(\frac{z+i}{1-i}\right)^2 - 4\left(\frac{z+i}{1-i}\right)^3 + \dots\right]$$

$$= \frac{i}{2} \left[1 - 2\left(\frac{z+i}{1-i}\right) + 3\left(\frac{z+i}{1-i}\right)^2 - 4\left(\frac{z+i}{1-i}\right)^3 + \dots\right]$$

Now radius, $R = \lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1} + 1}$

$$f(z) = \left[\frac{1}{(1-i)^2} - \frac{2(z+i)}{(1-i)^3} + \frac{3(z+i)^2}{(1-i)^4} - \dots \right]$$

Here $a_n = \frac{1}{(1-i)^{2+n}}$, $a_{n+1} = \frac{1}{(1-i)^{2+n+1}} = \frac{1}{(1-i)^{3+n}}$

$$R = \frac{a_n}{a_{n+1}} = \frac{1}{(1-i)^{2+n}} \times \frac{(1-i)^{2+n} \cdot (1-i)}{1} = (1-i), \quad |R| = \sqrt{2}$$

(g) $f(z) = \cos z$ about the point $z_0 = \frac{\pi}{2}$.

Solution: $f(z) = f(z_0) + (z-z_0)f'(z_0) + \frac{f''(z_0)}{2!} (z-z_0)^2 + \frac{f'''(z_0)}{3!} (z-z_0)^3 + \dots$

$$f(z_0)=0, \quad f'(z_0)=-1, \quad f''(z_0)=0, \quad f'''(z_0)=1, \quad f^{(4)}(z_0)=0, \quad f^{(5)}(z_0)=-1$$

$$f(z)=0+\left(z-\frac{\pi}{2}\right)(-1)+0+1\left(z-\frac{\pi}{2}\right)^3 \cdot \frac{1}{3!}+0+(-1)\left(z-\frac{\pi}{2}\right)^5 \cdot \frac{1}{5!}+\dots$$

$$f(z)=(-1)\left(z-\frac{\pi}{2}\right)+\frac{1}{3!}\left(z-\frac{\pi}{2}\right)^3-\left(z-\frac{\pi}{2}\right)^5 \cdot \frac{1}{5!}+\dots$$

$$a_1=-1, \quad a_n=(-1)^{1+n}, \quad a_{n+1}=(-1)^{n+2}, \quad R=\frac{(-1)^{n+1}}{(-1)^{n+2}}=-1, \quad |R|=1$$

(h) $f(z)=\frac{z-1}{z+1}$ about the point $z_0=1$

Solution: $f(z)=\frac{z-1}{z+1}=\frac{z-1}{1+(z-1)+1}=\frac{z-1}{2+(z-1)}=\frac{(z-1)}{2\left[1+\frac{z-1}{2}\right]}=\frac{z-1}{2}\left[1+\frac{z-1}{2}\right]^{-1}$

$$f(z)=\frac{z-1}{2}\left[1+\frac{z-1}{2}+\left(\frac{z-1}{2}\right)^2+\left(\frac{z-1}{2}\right)^3+\dots\right]$$

$$a_1=\frac{1}{2}, \quad a_n=\left(\frac{1}{2}\right)^{n+1}, \quad a_{n+1}=\left(\frac{1}{2}\right)^{n+2}, \quad R=\frac{a_n}{a_{n+1}}=\frac{1}{2}$$

(i) $f(z)=\frac{1}{1-z}$ about $z_0=i$.

Solution: $f(z)=\frac{1}{1-z} \Rightarrow \frac{1}{1-z}=\frac{1}{1-(z-i)-i}=\frac{1}{(1-i)-(z-i)}$

$$f(z)=\frac{1}{(1-i)\left[1-\frac{z-i}{1-i}\right]}=\frac{1}{1-i}\left[1-\frac{z-i}{1-i}\right]^{-1}$$

$$f(z)=\frac{1}{1-i}\left[1+\left(\frac{z-i}{1-i}\right)+\left(\frac{z-i}{1-i}\right)^2+\left(\frac{z-i}{1-i}\right)^3+\dots\right]$$

$$f(z)=\left(\frac{1}{1-i}\right)+\frac{(z-i)}{(1-i)^2}+\frac{(z-i)^2}{(1-i)^3}+\frac{(z-i)^3}{(1-i)^4}+\dots$$